Evaluating $\displaystyle\int_0^\infty \frac{\log x}{1+x^2}\,dx$ by the Method of Residues

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Source: L. V. Ahlfors, *Complex Analysis*, 3rd ed., Chapter 4, Section 5.3, Exercise 3(h) (MIT OCW 18.112, Problem Set 5)

Problem Statement. Evaluate $I = \displaystyle\int_0^\infty \frac{\log x}{1+x^2}\,dx$ by the method of residues.

Answer. $I = 0$.

Convergence

The integral converges absolutely. Near 0, $\left|\frac{\log x}{1+x^2}\right| \le |\log x|$, and $\int_0^1 |\log x|\,dx = 1$. Near ∞, $\left|\frac{\log x}{1+x^2}\right| \le \frac{\log x}{x^2}$ for $x \ge 1$, and $\int_1^\infty \frac{\log x}{x^2}\,dx = 1$. Hence I is a well-defined real number, and

$$I = \lim_{\substack{\rho\to 0^+ \\ R\to\infty}} \int_\rho^R \frac{\log x}{1+x^2}\,dx.$$

Setup

Choose the branch of the logarithm

$$\log z = \ln|z| + i\arg z, \qquad -\frac{\pi}{2} < \arg z < \frac{3\pi}{2}.$$

It is analytic on $\mathbb{C}$ minus the closed negative imaginary axis, a region that contains the closed upper half-plane except for 0. On the positive real axis it agrees with the real logarithm. On the negative real axis, writing $z = -x$ with $x > 0$, it gives $\log z = \ln x + i\pi$.

Let

$$f(z) = \frac{\log z}{1+z^2}.$$

In the upper half-plane, f is analytic except for a simple pole at $z = i$.

For $0 < \rho < 1 < R$, let γ be the closed contour, traversed counterclockwise, consisting of:

- the segment $[\rho, R]$ on the positive real axis;
- the large semicircle $C_R : z = Re^{i\theta}$, $0 \le \theta \le \pi$;
- the segment $[-R, -\rho]$ on the negative real axis;
- the small semicircle C_ρ of radius ρ about 0 in the upper half-plane, traversed clockwise from $-\rho$ to ρ.

The only singularity of f enclosed by γ is $z = i$.

The Residue

At $z = i$, the denominator factors as $1 + z^2 = (z - i)(z + i)$, so

$$\operatorname{Res}_{z=i} f = \frac{\log i}{2i} = \frac{i\pi/2}{2i} = \frac{\pi}{4},$$

since $\log i = \ln 1 + i \cdot \frac{\pi}{2} = \frac{i\pi}{2}$ for our branch. By the residue theorem,

$$\int_\gamma f(z)\,dz = 2\pi i \cdot \frac{\pi}{4} = \frac{i\pi^2}{2}. \tag{1}$$

The Pieces of the Contour

The real-axis segments. On $[\rho, R]$ we have $\int_\rho^R \frac{\ln x}{1+x^2}\,dx$. On $[-R, -\rho]$, substitute $z = -x$ with $\rho \le x \le R$. Then $dz = -dx$, the limits reverse, and $\log z = \ln x + i\pi$, so

$$\int_{-R}^{-\rho} \frac{\log z}{1+z^2}\,dz = \int_\rho^R \frac{\ln x + i\pi}{1+x^2}\,dx.$$

Together, the two segments contribute

$$2\int_\rho^R \frac{\ln x}{1+x^2}\,dx + i\pi \int_\rho^R \frac{dx}{1+x^2}. \tag{2}$$

The large semicircle. On C_R, $|\log z| \le \ln R + \pi$ and $|1 + z^2| \ge R^2 - 1$. The length of C_R is πR, so

$$\left| \int_{C_R} f(z)\,dz \right| \le \frac{\pi R(\ln R + \pi)}{R^2 - 1} \longrightarrow 0 \qquad \text{as } R \to \infty.$$

The small semicircle. On C_ρ, $|\log z| \le |\ln \rho| + \pi$ and $|1 + z^2| \ge 1 - \rho^2$. The length of C_ρ is $\pi\rho$, so

$$\left| \int_{C_\rho} f(z)\,dz \right| \le \frac{\pi\rho(|\ln \rho| + \pi)}{1 - \rho^2} \longrightarrow 0 \qquad \text{as } \rho \to 0^+,$$

since $\rho \ln \rho \to 0$ as $\rho \to 0^+$.

Conclusion

Combining (1) and (2) with the two semicircle contributions, and letting $\rho \to 0^+$ and $R \to \infty$, gives

$$2I + i\pi \int_0^\infty \frac{dx}{1+x^2} = \frac{i\pi^2}{2}.$$

Since I is real, comparing real parts gives $2I = 0$. Therefore

$$\int_0^\infty \frac{\log x}{1+x^2}\,dx = 0.$$

Comparing imaginary parts gives $\pi \int_0^\infty \frac{dx}{1+x^2} = \frac{\pi^2}{2}$, which recovers the familiar value $\int_0^\infty \frac{dx}{1+x^2} = \frac{\pi}{2}$ and serves as a check on the computation. □